$$\sqrt{\mathrm{MSE}} \propto s^{-\frac{\beta}{2\beta+1}}. \tag{6.48}$$

This is what we found in Section 1.1.3 as well.

These calculations show how the order β of a scheme affects both the optimal allocation of effort and the convergence rate under the optimal allocation. As the convergence order β increases, the optimal convergence rate in (6.48) approaches $s^{-1/2}$, the rate associated with unbiased simulation. But for the important cases of $\beta = 1$ (first-order) and $\beta = 2$ (second-order) we get rates of $s^{-1/3}$ and $s^{-2/5}$. This makes precise the notion that simulating a process for which a discretization scheme is necessary is harder than simulating a solvable model. It also shows that when very accurate results are required (i.e., when s is large), a higher-order scheme will ultimately dominate a lower-order scheme.

Duffie and Glynn [100] prove a limit theorem that justifies the convergence rate implied by (6.48). They also report numerical results that are generally consistent with their theoretical predictions.

6.4 Extremes and Barrier Crossings: Brownian Interpolation

In Section 6.2.3 we showed that discretization methods can sometimes be extended to path-dependent payoffs through supplementary state variables. The additional state variables remove dependence on the past; standard discretization procedures can then be applied to the augmented state vector.

In option pricing applications, path-dependence often enters through the maximum or minimum of an underlying asset over the life of the option. This includes, for example, options whose payoffs depend on whether or not an underlying asset crosses a barrier. Here, too, path-dependence can be eliminated by including the running maximum or minimum in the state vector. However, this renders standard discretization procedures inapplicable because of the singular dynamics of these supplementary variables. The running maximum, for example, can increase only when it is equal to the underlying process.

This issue arises even when the underlying process X is a standard Brownian motion. Let

$$M(t) = \max_{0 \le u \le t} X(u)$$

and let

$$\hat{M}^h(n) = \max\{X(0), X(h), X(2h), \ldots, X(nh)\}. \tag{6.49}$$

Then $\hat{M}^h(n)$ is the maximum of the Euler approximation to X over $[0, nh]$; the Euler approximation to X is exact for X itself because X is Brownian motion. Fix a time T and let $h = T/n$ so that $\hat{M}^h(n)$ is the discrete-time approximation of $M(T)$. Asmussen, Glynn, and Pitman [24] show that the normalized error

$$h^{-1/2}[\hat{M}^h(n) - M(T)]$$

has a limiting distribution as $h \to 0$. This result may be paraphrased as stating that the distribution of $\hat{M}^h(n)$ converges to that of $M(T)$ at rate $h^{1/2}$. It follows that the weak order of convergence (in the sense of (6.14)) cannot be greater than 1/2. In contrast, we noted in Section 6.1.2 that for SDEs with smooth coefficient functions the Euler scheme has weak order of convergence 1. Thus, the singularity of the dynamics of the running maximum leads to a slower convergence rate.

In the case of Brownian motion, this difficulty can be circumvented by sampling $M(T)$ directly, rather than through (6.49). We can sample from the joint distribution of $X(T)$ and $M(T)$ as follows. First we generate $X(T)$ from $N(0,T)$. Conditional on $X(T)$ the process $\{X(t), 0 \leq t \leq T\}$ becomes a Brownian bridge, so we need to sample from the distribution of the maximum of a Brownian bridge. We discussed how to do this in Example 2.2.3. Given $X(T)$, set

$$M(T) = \frac{X(T) + \sqrt{X(T)^2 - 2T \log U}}{2}$$

with $U \sim$ Unif[0,1] independent of $X(T)$. The pair $(X(T), M(T))$ then has the joint distribution of the terminal and maximum value of the Brownian motion over $[0,T]$.

This procedure, exact for Brownian motion, suggests an approximation for more general processes. Suppose X is a diffusion satisfying the SDE (6.1) with scalar coefficient functions a and b. Let $\hat{X}(i) = \hat{X}(ih)$, $i = 0, 1, \ldots$, be a discrete-time approximation to X, such as one defined through an Euler or higher-order scheme. The simple estimate (6.49) applied to $\hat{X}$ is equivalent to taking the maximum over a piecewise linear interpolation of $\hat{X}$. We can expect to get a better approximation by interpolating over the interval $[ih, (i+1)h)$ using a Brownian motion with fixed parameters $a_i = a(\hat{X}(i))$ and $b_i = b(\hat{X}(i))$. Given the endpoints $\hat{X}(i)$ and $\hat{X}((i+1))$, the maximum of the interpolating Brownian bridge can be simulated using

$$\hat{M}_i = \frac{\hat{X}(i+1) + \hat{X}(i) + \sqrt{[\hat{X}(i+1) - \hat{X}(i)]^2 - 2b_i^2 h \log U_i}}{2}, \tag{6.50}$$

with $U_0, U_1, \ldots$ independent Unif[0,1] random variables. (The value of $a(\hat{X}(i))$ becomes immaterial once we condition on $\hat{X}(i+1)$.) The maximum of X over $[0,T]$ can then be approximated using

$$\max\{\hat{M}_0, \hat{M}_1, \ldots, \hat{M}_{n-1}\}.$$

Similar ideas are suggested in Andersen and Brotherton-Ratcliffe [16] and in Beaglehole, Dybvig, and Zhou [41] for pricing lookback options; their numerical results indicate that the approach can be very effective. Baldi [31] analyzes related techniques in a much more general setting.

In some applications, X may be better approximated by geometric Brownian motion than by ordinary Brownian motion. This can be accommodated by applying (6.50) to $\log \hat{X}$ rather than $\hat{X}$. This yields

$$\log \hat{M}_i = \frac{\log(\hat{X}(i+1)\hat{X}(i)) + \sqrt{[\log(\hat{X}(i+1)/\hat{X}(i))]^2 - 2(b_i/\hat{X}(i))^2 h \log U_i}}{2},$$

and exponentiating produces $\hat{M}_i$.

Barrier Crossings

Similar ideas apply in pricing barrier options with continuously monitored barriers. Suppose $B > X(0)$ and let

$$\tau = \inf\{t \geq 0 : X(t) > B\}.$$

A knock-out option might have a payoff of the form

$$(K - X(T))^+ \mathbf{1}\{\tau > T\}, \tag{6.51}$$

with K a constant. This requires simulation of $X(T)$ and the indicator $\mathbf{1}\{\tau > T\}$.

The simplest method sets

$$\hat{\tau} = \inf\{i : \hat{X}(i) > B\}$$

and approximates $(X(T), \mathbf{1}\{\tau > T\})$ by $(\hat{X}(n), \mathbf{1}\{\hat{\tau} > n\})$ with $h = T/n$, for some discretizaton $\hat{X}$. But even if we could simulate X exactly on the discrete grid $0, h, 2h, \ldots$, this would not sample $\mathbf{1}\{\tau > T\}$ exactly: it is possible for X to cross the barrier at some time t between grid points ih and $(i+1)h$ and never be above the barrier at any of the dates $0, h, 2h, \ldots$.

The method in (6.50) can be used to reduce discretization error in sampling the survival indicator $\mathbf{1}\{\tau > T\}$. Observe that the barrier is crossed in the interval $[ih, (i+1)h)$ precisely if the maximum over this interval exceeds B. Hence, we can approximate the survival indicator $\mathbf{1}\{\tau > T\}$ using

$$\prod_{i=0}^{n-1} \mathbf{1}\{\hat{M}_i \leq B\}, \tag{6.52}$$

with $nh = T$ and $\hat{M}_i$ as in (6.50).

This method can be simplified. Rather than generate $\hat{M}_i$, we can sample the indicators $\mathbf{1}\{\hat{M}_i \leq B\}$ directly. Given $\hat{X}(i)$ and $\hat{X}(i+1)$, this indicator takes the value 1 with probability

$$\hat{p}_i = P(\hat{M}_i \leq B | \hat{X}(i), \hat{X}(i+1)) = 1 - \exp\left(-\frac{2(B - \hat{X}(i))(B - \hat{X}(i+1))}{b(\hat{X}(i))^2 h}\right),$$

(assuming B is greater than both $\hat{X}(i)$ and $\hat{X}(i+1)$) and it takes the value 0 with probability $1-\hat{p}_i$. Thus, we can approximate $\mathbf{1}\{\tau > T\}$ using

$$\prod_{i=0}^{n-1} \mathbf{1}\{U_i \le \hat{p}_i\}.$$

For fixed $U_0, U_1, \ldots, U_{n-1}$, this has the same value as (6.52) but is slightly simpler to evaluate. The probabilities $\hat{p}_i$ could alternatively be computed based on an approximating geometric (rather than ordinary) Brownian motion.

The discretized process $\hat{X}$ is often a Markov process and this leads to further simplification. Consider, for example, the payoff in (6.51). Using (6.52), we approximate the payoff as

$$(K - \hat{X}(n))^+ \prod_{i=0}^{n-1} \mathbf{1}\{\hat{M}_i \le B\}. \tag{6.53}$$

The conditional expectation of this expression given the values of $\hat{X}$ is

$$\mathsf{E}\left[(K - \hat{X}(n))^+ \prod_{i=0}^{n-1} \mathbf{1}\{\hat{M}_i \le B\} | \hat{X}(0), \hat{X}(1), \ldots, \hat{X}(n)\right] \tag{6.54}$$

$$= (K - \hat{X}(n))^+ \prod_{i=0}^{n-1} \mathsf{E}[\mathbf{1}\{\hat{M}_i \le B\} | \hat{X}(i), \hat{X}(i+1)]$$

$$= (K - \hat{X}(n))^+ \prod_{i=0}^{n-1} \hat{p}_i. \tag{6.55}$$

Thus, rather than generate the barrier-crossing indicators, we can just multiply by the probabilities $\hat{p}_i$.

Because (6.55) is the conditional expectation of (6.53), the two have the same expectation and thus the same discretization bias. By Jensen's inequality, the second moment of (6.53) is larger than the second moment of its conditional expectation (6.55), so using (6.55) rather than (6.53) reduces variance. (This is an instance of a more general strategy for reducing variance known as *conditional Monte Carlo*, based on replacing an estimator with its conditional expectation; see, Boyle et al. [53] for other applications in finance.) Using (6.53), we would stop simulating a path once some $\hat{M}_i$ exceeds B. Using (6.55), we never generate the $\hat{M}_i$ and must therefore simulate every path for n steps, unless some $\hat{X}(i)$ exceeds B (in which case $\hat{p}_i = 0$). So, although (6.55) has lower variance, it requires greater computational effort per path. A closely related tradeoff is investigated by Glasserman and Staum [146]; they consider estimators in which each transition of an underlying asset is sampled conditional on not crossing a barrier. In their setting, products of survival probabilities like those in (6.55) serve as likelihood ratios relating the conditional and unconditional evolution of the process.

Baldi, Caramellino, and Iovino [32] develop methods for reducing discretization error in a general class of barrier option simulation problems. They consider single- and double-barrier options with time-varying barriers and develop approximations to the one-step survival probabilities that refine the $\hat{p}_i$ above. The $\hat{p}_i$ are based on a single constant barrier and a Brownian approximation over a time interval of length h. Baldi et al. [32] derive asymptotics of the survival probabilities as $h \to 0$ for quite general diffusions based, in part, on a linear approximation to upper and lower barriers.

Averages Revisited

As already noted, the simulation estimators based on (6.50) or (6.52) can be viewed as the result of using Brownian motion to interpolate between the points $\hat{X}(i)$ and $\hat{X}(i+1)$ in a discretization scheme. The same idea can be applied in simulating other path-dependent quantities besides extremes and barrier-crossing indicators.

As an example, consider simulation of the pair

$$\left(X(T), \int_0^T X(t)\,dt \right)$$

for some scalar diffusion X. In Section 6.2.3, we suggested treating this pair as the state at time T of a bivariate diffusion and applying a discretization method to this augmented process. An alternative simulates a discretization $\hat{X}(i)$, $i = 0, 1, \ldots, n$, and uses Brownian interpolation to approximate the integral. More explicitly, the approximation is

$$\int_0^T X(t)\,dt \approx \sum_{i=0}^{n-1} b_i \hat{A}_i,$$

with $b_i = b(\hat{X}(i))$ and each $\hat{A}_i$ sampled from the distribution of

$$\int_t^{t+h} W(u)\,du, \quad W \sim \mathrm{BM}(0,1),$$

conditional on $W(t) = \hat{X}(i)$ and $W(t+h) = \hat{X}(i+1)$. The calculations used to derive (6.27) show that this conditional distribution is normal with mean $h(\hat{X}(i+1) + \hat{X}(i))/2$ and variance $h^3/3$. Thus, the $\hat{A}_i$ are easily generated.

This leads to a discretization scheme only slightly different from the one arrived at through the approach in Section 6.2.3. Because of the relative smoothness of the running integral, the effect of Brownian interpolation in this setting is minor compared to the benefit in simulating extremes or barrier crossings.